PHYSICS

1) An air bubble in a glass slab with refractive index 1.5 (near normal incidence) is 5 cm deep when viewed from one surface and 5 cm deep when viewed from the opposite face. The thickness (in cm) of the slab is :-

(1) 12
(2) 15
(3) 8
(4) 10

2) A ray of light is incident at an angle of 60° on one face of a prism of angle 30°. The emergent ray of light makes an angle of 30° with incident ray. The angle made by emergent ray with second face of prism will be :-

(1) 0°
(2) 90°
(3) 45°
(4) 30°

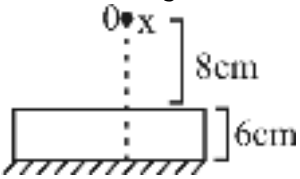
3) When monochromatic red light is used instead of blue light in a convex lens, its focal length will :

(1) Increase
(2) Decrease
(3) Remain same
(4) Does not depend on colour of light

4) An object is placed at a distance of 10 cm from the first focus of a convex lens when real image is formed at a distance of 40 cm from the second focus. Then the focal length of lens is -

(1) 20 cm
(2) 15 cm
(3) 8 cm
(4) 13.33 cm

5) An object is placed at 8 cm from the upper face of a glass slab of thickness 6 cm. The lower face of the slab is silvered as shown in the figure. If the refractive index of the material of slab is 1.5 find the position of the image :-



(1) 6 cm behind the silvered face

(2) 8 cm behind the silvered face

(3) 10 cm behind the silvered face

(4) 4 cm behind the silvered face

6) Diameter of the moon is 3.5×10^3 km and its distance from earth is 3.8×10^5 km. It is seen by a telescope whose objective and eyepiece have focal length 4 m and 10 cm. What will the angular diameter of the image of the moon.

(1) 0.36 radian

(2) 0.20 radian

(3) 0.46 radian

(4) 0.50 radian

7) A single slit of width 0.20 mm is illuminated with light of wavelength 500nm.The observing screen is placed 80 cm from the slit. The width of the central bright fringe will be :

(1) 1 mm

(2) 2 mm

(3) 4 mm

(4) 5 mm

8) A polarized light of intensity I_0 is passed through another polarizer whose pass axis makes an angle of 60° with the pass axis of the former. What is the intensity of emergent polarized light from second polarizer ?

(1) $I = \frac{I_0}{5}$

(2) $I = I_0$

(3) $I = \frac{I_0}{4}$

(4) $I = \frac{I_0}{6}$

9) A beam of light strikes a piece of glass at an angle of incidence of 60° and the reflected beam is completely plane polarised. The refractive index of the glass is :-

(1) 1.5

(2) $\sqrt{3}$

(3) $\sqrt{2}$

(4) (3/2)

10) **Assertion :** Diffraction occur in sound but do not occur in light waves. **Reason :** Wavelength of light wave is less than the wavelength of sound.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

11) If diffraction occurs through a single slit then intensity of second maxima become ..... % of central maxima :-

(1) 4 %

(2) 25 %

(3) 1.64 %

(4) 50 %

12) Plane polarised light is passed through a polaroid. On viewing through the polariod we find that when the polariod is given one complete rotation about the direction of the light, one of the following is observed.

(1) The intensity of light gradually decreases to zero and remains at zero

(2) The intensity of light gradually increases to a maximum and remains at maximum

(3) There is no change in intensity

(4) The intensity of light is twice maximum and twice zero

13) In a double slit experiment $\lambda = 405$ nm, the slit separation is 20 µm and slit width is 4 µm. Number of bright fringes within central peak of diffraction envelop is :-

(1) 6

(2) 8

(3) 10

(4) 12

14) Four polaroids are placed such that the optic axis of each is inclined at an angle of 30° with the optic axis of the preceding one. If unpolarised light of intensity I_0 falls on the first polaroid, the light transmitted from the fourth is :-

(1) I_0

(2) $I_0/16$

(3) $0.21\ I_0$

(4) $\frac{I_0}{2}$

15) A single slit of width b is illuminated by a coherent monochromatic light of wavelength λ. If the second and fourth minima in the diffraction pattern at a distance 1 m from the slit are at 3 cm and 6 cm respectively from the central maximum, what is the width of the central maximum? (i.e., distance between first minimum on either side of the central maximum)

(1) 4.5 cm

(2) 1.5 cm

(3) 6.0 cm

(4) 3.0 cm

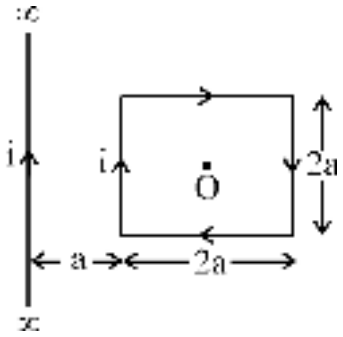
16) Uniform circular motion of a planet around a sun is which type of motion

(1) Periodic and oscillatory both motions
(2) Periodic motion but not oscillatory motion
(3) oscillatory motion but not periodic motion
(4) None of these

17) ***Assertion :-*** In SHM acceleration is always directed towards the mean position.
Reason :- The body in SHM stops momentary at the extreme positions and then moves back to mean position.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false
(4) Assertion is false but Reason is true.

18) Find net magnetic field at centre of square.



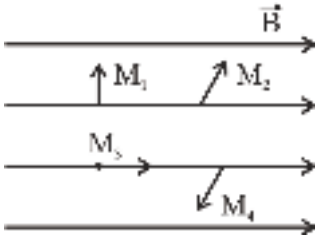
(1) $\frac{\mu_0 i}{2\pi a}\left[\frac{1}{2}+2\sqrt{2}\right]$
(2) $\frac{\mu_0 i}{2\pi a}\left[\frac{1}{2}+4\sqrt{2}\right]$
(3) $\frac{\mu_0 i}{2\pi a}\left[1+2\sqrt{2}\right]$
(4) $\frac{\mu_0 i}{2\pi a}\left[1-2\sqrt{2}\right]$

19) There are four light-weight rod samples A,B,C D separately suspended by threads. A bar magnet is slowly brought near each sample and the following observations are noted :-
(i) A is feebly repelled
(ii) B is feebly attracted
(iii) C is strongly attracted
(iv) D remains unaffected
Which one of the following is true ?

(1) B is of a paramagnetic material
(2) C is of a diamagnetic material
(3) D is of a ferromagnetic material
(4) A is of a non-magnetic material

20) The magnetic moment of a magnetic needle is M. It is placed in a uniform magnetic field B in

four different orientation named by M_1, M_2, M_3 and M_4. If potential energies in these orientation are



U_1, U_2, U_3 and U_4 respectively, then : -

(1) $U_4 < U_2 < U_3 < U_1$

(2) $U_2 < U_3 < U_4 < U_1$

(3) $U_3 < U_2 < U_1 < U_4$

(4) $U_1 < U_2 < U_3 < U_4$

21) Magnets A and B are geometrically similar but the magnetic moment of A is twice that of B. If T_1 and T_2 be the time periods of the oscillation in external magnetic field when their like poles and unlike poles are kept together respectively, then $\frac{T_1}{T_2}$ will be

(1) $\frac{1}{3}$

(2) $\frac{1}{2}$

(3) $\frac{1}{\sqrt{3}}$

(4) $\sqrt{3}$

22) A short bar magnet placed with its axis at 30° with a uniform external magnetic field of 0.16 T experiences a torque of magnitude 0.032 Nm. The magnetic moment of the bar magnet will be

(1) $\frac{0.23}{\frac{Nm}{T}}$

(2) $\frac{0.40}{\frac{Nm}{T}}$

(3) $\frac{0.80}{\frac{Nm}{T}}$

(4) Zero

23) Due to the flow of current in a circular loop of radius R, the magnetic induction produced at the centre of the loop is B. The magnetic moment of the loop is :- (μ_0 = permeability constant)

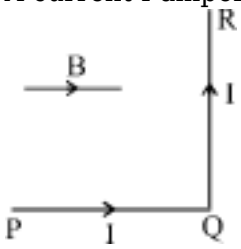
(1) $BR^3/2\pi\mu_0$

(2) $2\pi BR^3/\mu_0$

(3) $BR^2/2\pi\mu_0$

(4) $2\pi BR^2/\mu_0$

24) A wire PQR is bent as shown in figure and is placed in a region of uniform magnetic field B. The length of PQ = QR = □. A current I ampere flows through the wire as shown. The magnitude of force on PQ and QR will be :-



(1) BI□, 0

(2) 2 Bi□, 0

(3) 0, Bi□

(4) 0, 0

25) In a hydrogen atom, an electron moves in a circular orbit of radius 5.2×10^{-11} m and produces a magnetic induction of 12.56 T at its nucleus. The current produced by the motion of the electron will be (Given $\mu_0 = 4\pi \times 10^{-7}$ Wb/A–m)

(1) 6.53×10^{-3} ampere

(2) 13.25×10^{-10} ampere

(3) 9.6×10^{6} ampere

(4) 1.04×10^{-3} ampere

26) A beam of well collimated cathode rays travelling with a speed of 5×10^{6} ms^{-1} enter a region of mutually perpendicular electric and magnetic fields and emerge undeviated from this region. If $|B| = 0.02T$, the magnitude of the electric field is

(1) $10^{5} Vm^{-1}$

(2) $2.5 \times 10^{8} Vm^{-1}$

(3) $1.25 \times 10^{10} Vm^{-1}$

(4) $2 \times 10^{3} Vm^{-1}$

27) Ampere's law is used to calculate magnetic field due to

(1) Inside current only

(2) Outside current only

(3) Both inside and outside currents

(4) None of these

28) Two toroids 1 and 2 have total number of turns 200 and 100 respectively with average radii 40 cm and 80 cm respectively. If they carry same current i, then the ratio of the magnetic fields along the two is :

(1) 1 : 1

(2) 4 : 1

(3) 2 : 1

(4) 1 : 2

29) A sniper fires a rifle bullet into a gasoline tank making a hole 75 m below the surface of gasoline. The tank was sealed at 4.3 atm. The stored gasoline has a density of 660 kgm^{-3}. The velocity with which gasoline begins to shoot out of the hole is :-

(1) 25 ms^{-1}

(2) 50 ms^{-1}

(3) 75 ms^{-1}

(4) 100 ms^{-1}

30) Water is entering in a cylindrical container open from top at the rate of α container is consist of an orifice of area A at base. Find maximum height upto which water can be filled :

(1) $\frac{\alpha}{2gA}$

(2) $\frac{\alpha^2}{2gA^2}$

(3) $\frac{\alpha^2}{2gA}$

(4) $\frac{\alpha}{2gA^2}$

31) A tank is filled to a height H. The range of water coming out of a hole which is a depth $\frac{H}{4}$ from the surface of water level is:

(1) $\frac{2H}{\sqrt{3}}$

(2) $\frac{\sqrt{3}H}{2}$

(3) $\sqrt{3}H$

(4) $\frac{3H}{4}$

32) Water is flowing streamline motion through a horizontal tube. The pressure at a point in a tube is P where the velocity of flow is v. At another point where the pressure is $\frac{3P}{4}$, the velocity of flow is (density of water = ρ) :-

(1) $\sqrt{v^2+\frac{P}{\rho}}$

(2) $\sqrt{v^2+\frac{P}{2\rho}}$

(3) $\sqrt{v^2+\frac{2P}{\rho}}$

(4) $\sqrt{v^2 - \frac{2P}{\rho}}$

33) Two metal spheres are falling through a liquid of density 2×10^3 kg/m^3 with the same uniform speed. The material density of sphere 1 and sphere 2 are 8×10^3 kg/m^3 and 11×10^3 kg/m^3 respectively. The ratio of their radii is :

(1) $\frac{11}{8}$

(2) $\sqrt{\frac{11}{8}}$

(3) $\frac{3}{2}$

(4) $\sqrt{\frac{3}{2}}$

34) Two rain drops reach the earth with different terminal velocity having ratio 9 : 4 then the ratio of their volume is :

(1) 3 : 2

(2) 4 : 9

(3) 9 : 4

(4) 27 : 8

35) A layer of glycerine of thickness 1 mm is present between a large surface and small surface of area 0.1m^2. With what force the small surface is to be pulled, so that it can move with a velocity of 1 m/s:
(coefficient of viscosity = 0.07kg-m^{-1}s^{-1})

(1) 70N

(2) 7N

(3) 700 N

(4) 0.70N

36) **Assertion (A) :** A thin stainless steel needle can lie floating on a still water surface.
Reason (R) : When the buoyancy force balances the weight of the object, object floats.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is incorrect but (R) is correct

(4) Both (A) and (R) are correct but (R) is the correct explanation of (A)

37) A thin light wire is bent in the form of a ring of radius 2 cm. The ring is placed horizontally on the surface of soap solution and then lift up slowly. The upward force required to lift the ring from surface of soap solution is

(Surface tension = 3×10^{-2} N m^{-1})

(1) 5 mN

(2) 7.5 mN

(3) 8 mN

(4) 3.5 mN

38) The surface tension of soap solution is 0.03 N/m. The work done in blowing to form a soap bubble of surface area 40 cm^2 is:

(1) 1.2×10^{-4} J

(2) 2.4×10^{-4} J

(3) 12×10^{-4} J

(4) 24×10^{-4} J

39) The surface tension of a soap solution is 25×10^{-3} Nm^{-1}. The excess pressure inside a soap bubble of diameter 1cm is

(1) 10 Pa

(2) 20 Pa

(3) 5 Pa

(4) None of the above

40) Three photo diodes D_1, D_2 and D_3 are made of semiconductors having band gap of 2eV, 2.5 eV and 3eV respectively which one will be able to detect light of wave length 6000 Å :

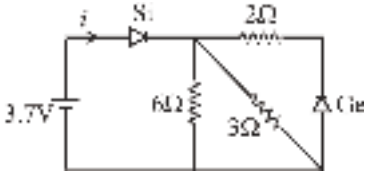
(1) D_1

(2) D_2

(3) D_3

(4) D_1 and D_2 both

41) Find current 'i' :



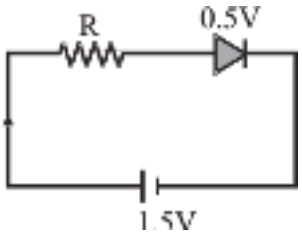
(1) 2A

(2) 3A

(3) 1.5A

(4) 0.5A

42) The diode used in the circuit shown in the figure has a constant voltage drop of 0.5 V at all currents and a maximum power rating of 100 milliwatts. What should be the value of the resistor R,



connected in series with the diode for obtaining maximum current:-

(1) 1.5 Ω

(2) 5 Ω

(3) 6.67 Ω

(4) 200 Ω

43) In an intrinsic semiconductor, the density of conduction electrons is $7.07 \times 10^{15}\ m^{-3}$. When it is doped with indium, the density of holes becomes $5 \times 10^{22}\ m^{-3}$. Find the density of conduction electrons in doped semiconductor :

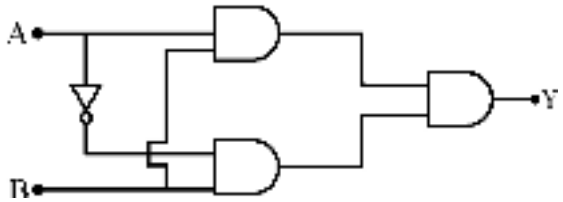
(1) Zero

(2) $1 \times 10^{9}\ m^{-3}$

(3) $7 \times 10^{15}\ m^{-3}$

(4) $5 \times 10^{22}\ m^{-3}$

44) For given circuit, truth table is :-



(1)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	0

(2)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

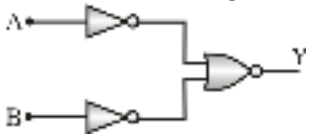
(3)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(4)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

45) The logic circuit shown in figure has the input wave forms A and B as shown. The correct output wave form is :-





(1) 

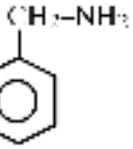
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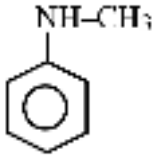
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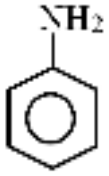
(4) None of these

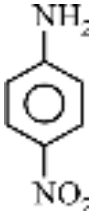
CHEMISTRY

1) Which of the following is the strongest base :-

(1) 

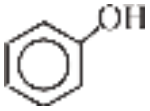
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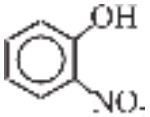
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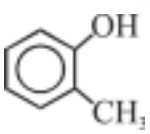
(4) 

2) Which one of the following compounds is most acidic ?

(1) $Cl–CH_2–CH_2–OH$

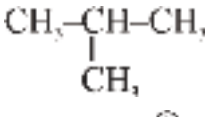
(2) 

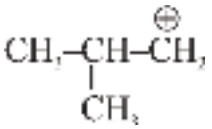
(3) 

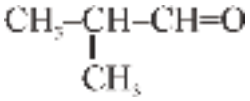
(4) 

3) In which of the molecule +I effect of $–CH_3$ group not present ?

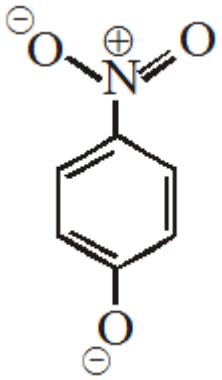
(1) $CH_2=CH–CH_3$

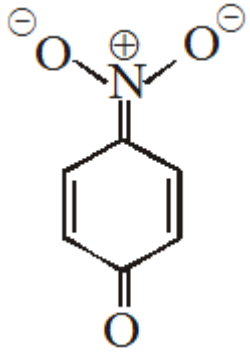
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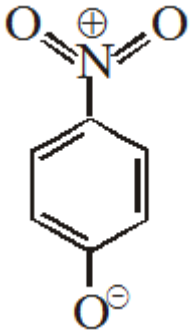
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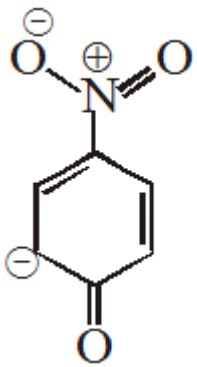
(4) 

4) The most unlikely representation of resonance structure of P-nitrophenoxide ion is :-

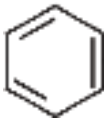
(1) 

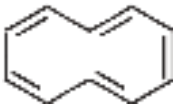
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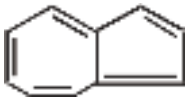
(3) 

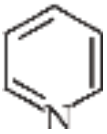
(4) 

5) Which of the following is/are aromatic hydrocarbon having $10\pi e^-$:-

(1) 

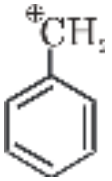
(2) 

(3) 

(4) 

6) Correct order of stability of carbocation given below is :-

(i) 

(ii) 

(iii) $CH_2 = CH-\overset{\oplus}{C}H_2$

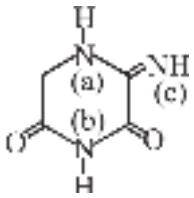
(iv) $CH_2 = \overset{\oplus}{C}H$

(1) i > ii > iii > iv

(2) ii > iii > i > iv

(3) ii > i > iii > iv

(4) i > iii > ii > iv

7) Which nitrogen is most basic :- 

(1) a

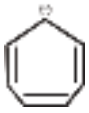
(2) b

(3) c

(4) a = b = c

8) In which of the following all C–C bond lengths are not equal?

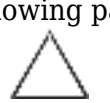
(1) 

(2) 

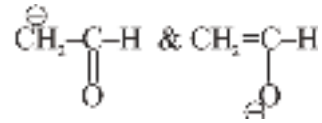
(3) 

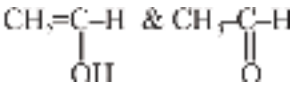
(4) 

9) Which of the following pairs are resonating structures:-

(I) $CH_2=CH-CH_3$ & 

(II) CH_3-O-CH_3 & CH_3-CH_2-OH

(III) 

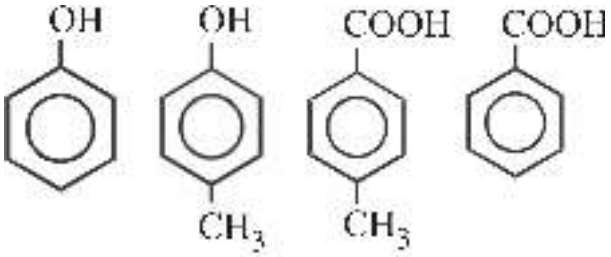
(IV) 

(1) I & II only

(2) III & IV only

(3) III only

(4) All of these



10) (a) (b) (c) (d)

Compare acidic strength

(1) a > c > d > b

(2) d > a > c > b

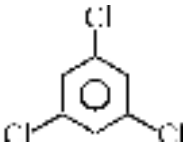
(3) d > c > a > b

(4) d > c > b > a

11) Which of the following has non-zero dipole moment?

(1) 

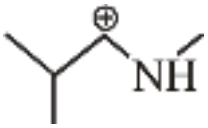
(2) 

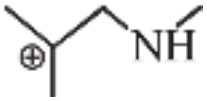
(3) 

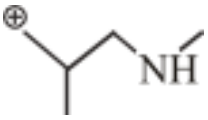
(4) 

12) Which is most stable :

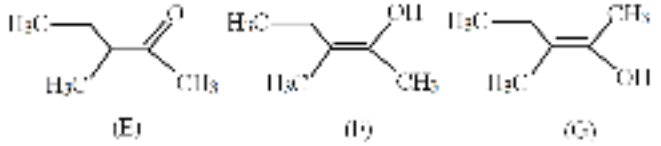
(1) 

(2) 

(3) 

(4) 

13) Which statement is incorrect for the structures E, F and G :



(1) E, F and G are tautomers

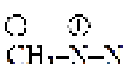
(2) F and G are geometrical isomers

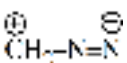
(3) E, F and G are resonance structures

(4) F and G are diastereomers

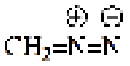
14)

The least stable resonating structure of diazomethane (CH_2N_2) is -

(1) 

(2) 

(3) 

(4) 

15) Which statement is correct for following compound?

(I)  (II) 

(1) I & II are aromatic

(2) I & II are antiaromatic

(3) I is aromatic and II is antiaromatic

(4) I is non-aromatic and II is anti-aromatic

16) Correct order of K_b in aqueous solution of the following compound :
Et_2NH ; Et_3N ; $EtNH_2$; NH_3

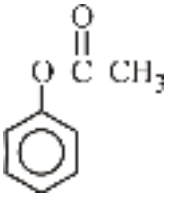
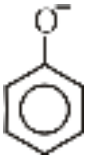
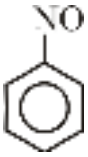
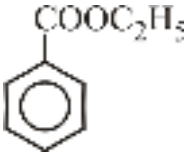
(1) $Et_2NH > EtNH_2 > Et_3N > NH_3$

(2) $Et_2NH > Et_3N > EtNH_2 > NH_3$

(3) $EtNH_2 > Et_2NH > Et_3N > NH_3$

(4) $Et_3N > Et_2NH > EtNH_2 > NH_3$

17) Match the column :-

Column-I (Compound)		**Column-II (Effect of substituent)**	
(a)		(p)	+M and +I
(b)		(q)	+M and –M
(c)		(r)	–M and –I
(d)		(s)	+M and –I

(1) a = s, b = q, c = r, d = p

(2) a = s, b = p, c = r, d = q

(3) a = s, b = p, c = q, d = r

(4) a = r, b = q, c = p, d = s

18)

Consider the following equilibrium in a closed container
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

At a fixed temperature, the volume of a reaction container is halved. For this change, which of the following statements holds true regarding the equilibrium constant (K_p) and degree of dissociation (α)?

(1) Neither K_P nor α changes

(2) Both K_P and α change

(3) K_P changes but α does not change

(4) K_P does not change but α changes

19) $C(s) + O_2(g) \rightleftharpoons CO_2(g)$,

The given reaction is started with 1 mol C, 4 mol O_2 and 3 mol CO_2. If $K_p = \frac{1}{5}$ then moles of CO_2 at equilibrium will be :-

(1) 2.31

(2) 1.17

(3) 5.28

(4) 8.29

20) Assuming the reaction $A_2(g) + B_2(g) \rightleftharpoons C(g)$ to be exothermic, the yield of the product (C) increases with :-
(a) Increase of temperature
(b) Increase of pressure
(c) Addition of catalyst
(d) Addition of inert gas at constant volume
(e) Removal of (C)

(1) a, b, c, d

(2) Only b

(3) b & e

(4) a & e

21) For this reaction $ax_{(g)} + by_{(g)} \rightleftharpoons bz_{(g)} + aw_{(g)}$

(1) $K_p = K_c(RT)^{a+b}$

(2) $K_p = K_c$

(3) $K_p = K_c\,(RT)$

(4) $K_p = \frac{K_c}{(a+b)^2}$

22) What is equilibrium constant for the reaction
$P_{4(s)} + 5O_{2(g)} \rightleftharpoons P_4O_{10(s)}$:-

(1) $K_c = \frac{[P_4O_{10}]}{[P_4][O_2]^5}$

(2) $K_c = \frac{[P_4O_{10}]}{5[P_4][O_2]^5}$

(3) $K_c = [O_2]^5$

(4) $K_c = \frac{1}{[O_2]^5}$

23) At 25°C the value of K_C for the reaction
$1/2N_2 + O_2 \rightleftharpoons NO_2$ is 2×10^4. What will be the value of K_C for $2NO_2 \rightleftharpoons N_2 + 2O_2$?

(1) 2×10^4

(2) $\frac{1}{4 \times 10^8}$

(3) 2×10^2

(4) 10^4

24) For the reaction : $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, the degree of dissociation in terms of K_P can be expressed as :-

(1) $\frac{K_P/P}{4 + K_P/P}$

(2) $\frac{K_P}{4 + K_P}$

(3) $\left[\frac{K_P/P}{4 + K_P/P}\right]^{1/2}$

(4) $\left[\frac{K_P}{4 + K_P}\right]^{1/3}$

25) Which is most stable oxide :-

(1) $2NO_2 \rightleftharpoons N_2 + 2O_2$ $K = 6.7 \times 10^{16}$

(2) $N_2O_2 \rightleftharpoons N_2 + O_2$ $K = 2.2 \times 10^{30}$

(3) $2N_2O_3 \rightleftharpoons 2N_2 + 3O_2$ $K = 1.2 \times 10^{34}$

(4) $2N_2O \rightleftharpoons 2N_2 + O_2$ $K = 3.5 \times 10^{33}$

26) For the given equilibrium at 500 K,
$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$
The value of equilibrium constant, K_p is 3.6×10^{-2}. Calculate the value of K_C for the reaction at this temperature ?
(R = 0.083 L-bar mol^{-1} K^{-1})

(1) 38

(2) 22

(3) 62

(4) 88

27) In the following equilibrium :–
I. $A + 2B \rightleftharpoons C$; K_1
II. $C + D \rightleftharpoons 3A$; K_2
III. $6B + D \rightleftharpoons 2C$; K_3 hence :-

(1) $3K_1 \times K_2 = K_3$

(2) $K_1^3 \times K_2^2 = K_3$

(3) $3K_1 \times K_2^2 = K_3$

(4) $K_1^3 \times K_2 = K_3$

28) For the decomposition of the compound, represented as
$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$
the $K_P = 2.9 \times 10^{-5}$ atm^3.
If the reaction is started with 1 mol of the compound, the total pressure at equilibrium would be

(1) 38.8×10^{-5} atm

(2) 1.94×10^{-2} atm

(3) 5.82×10^{-2} atm

(4) 7.66×10^{-3} atm

29) Which of the following factors will be favourable for higher yields of NO in the reaction given below?
$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

(1) Low temperature, high pressure and high concentration of N_2 and O_2

(2) High temperature and high concentration of N_2 and O_2. The reaction remains unaffected by pressure.

(3) High temperature, low pressure and low concentration of N_2 and O_2

(4) Low temperature, high concentrations of N_2 and O_2 and reaction remains unaffected by pressure.

30) XY_2 dissociates as :
$XY_2(g) \rightleftharpoons XY(g) + Y(g)$
Initial pressure of XY_2 is 600 mm Hg. The total pressure at equilibrium is 800 mm Hg. Assuming volume of system to remain constant, the value of K_p is : (in mm)

(1) 50

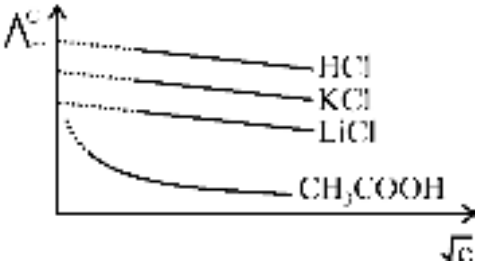
(2) 100

(3) 200

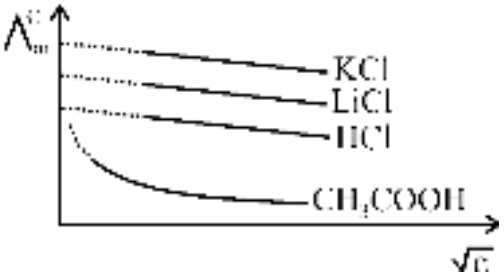
(4) 400

31) Select the correct option for following aqueous solutions :

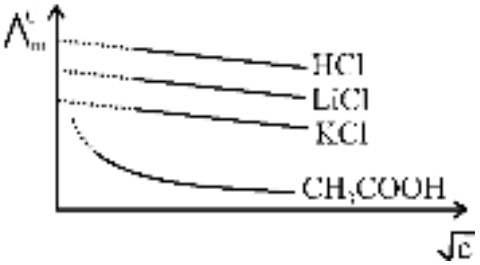
(1)



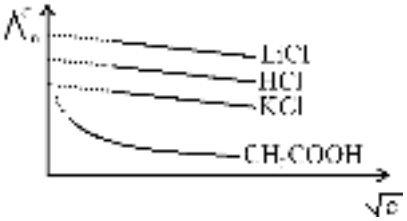
(2)



(3)



(4)



32) (i) $Cu + 2HCl \rightarrow CuCl_2 + H_2(g)$;

$E^{o}_{Cu^{2+}/Cu} = +0.34$ V

(ii) $Zn + 2HCl \rightarrow ZnCl_2 + H_2(g)$;

$E^{o}_{Zn^{2+}/Zn} = -0.76$ V

(iii) $Ag + 2HCl \rightarrow AgCl + H_2(g)$;

$E^{o}_{Ag^{+}/Ag} = +0.80$ V

Which of the above reactions is feasible ?

(1) (ii)

(2) (i)

(3) (iii)

(4) All of the above

33) 50 mL of 0.1 M $Ni(NO_3)_2$ solution is electrolysed using Ni electrode by passing 2×10^{-3} Faraday charge. The millimoles of $Ni(NO_3)_2$ left in solution are :-

(1) 4

(2) 6

(3) 8

(4) 5

34) During the electrolysis of $Ni(NO_3)_2$ solution, water is electrolysed and oxygen is liberated as per the equation :-
$2H_2O \rightarrow 4H^+ + O_2 + 4e^-$
when 3 ampere of current is passed for 965 seconds what is the volume of O_2 liberated at STP :-

(1) 0.168 L

(2) 11.2 L

(3) 5.6 L

(4) 0.112 L

35) The electrical resistance of a column of 0.05 mole/litre NaOH solution of diameter 1 cm and length 50 cm is 5.55×10^3 ohm. Calculate its molar conductivity :-

(1) 229.6 $Scm^2\ mol^{-1}$

(2) 119.5 $Scm^2\ mol^{-1}$

(3) 832 $Scm^2\ mol^{-1}$

(4) 420 $Scm^2\ mol^{-1}$

36) Statements:
(i) Unit of specific conductivity is $ohm^{-1}\ cm^{-1}$
(ii) Specific conductivity of strong electrolytes decreases on dilution
(iii) The amount of an ion discharged during electrolysis does not direclty depend upon resistance
(iv) The unit of electrochemical equivalent is g/coulomb

(1) All are correct

(2) All are wrong

(3) only (i), (ii) and (iv) are correct

(4) only (ii), (iii) and (iv) are correct

37) Ionisation constant of a weak acid (HA) in terms of Λ_m^∞ and Λ_m is :-

(1) $K_a = \frac{C\Lambda_m^\infty}{(\Lambda_m - \Lambda^\infty)}$

(2) $K_a = \frac{C\Lambda_m^2}{\Lambda_m^\infty(\Lambda_m^\infty - \Lambda_m)}$

(3) $K_a = \frac{C(\Lambda_m^\infty)^2}{\Lambda_m^\infty(\Lambda_m^\infty - \Lambda_m)}$

(4) $K_a = C\alpha^2$

38) $\Lambda_m^\infty(H_2CO_3)$ is equal to :

(1) $2\Lambda_m^\infty(HCl) + \Lambda_m^\infty(Na_2CO_3) - \Lambda_m^\infty(NaCl)$

(2) $\Lambda_m^\infty(HCl) + \Lambda_m^\infty(Na_2CO_3) - \Lambda_m^\infty(NaCl)$

(3) $2\Lambda_m^\infty(HCl) + \Lambda_m^\infty(Na_2CO_3) - 2\Lambda_m^\infty(NaCl)$

(4) $\Lambda_m^\infty (HCl) + \Lambda_m^\infty (Na_2CO_3) - 2\Lambda_m^\infty (NaCl)$

39) If the standard reduction potential E° for four divalent metals x, y, z, w are –1.46V, –0·36 V, 0·15 V and –1·24 V respectively, then -

(1) X will replace Z^{+2} from aqueous solution

(2) Y will replace Z^{+2} from aqueous solution

(3) W will replace Z^{+2} from aqueous solution

(4) All the above statements are correct

40) How many mole of $FeSO_4$, $H_2C_2O_4$ and FeC_2O_4 are oxidised separately by one mole of $KMnO_4$ in acidic medium respectively :-

(1) 5 ; 5/2 ; 5/3

(2) 2/5 ; 2 ; 5

(3) 5/3 ; 3/5 ; 2/5

(4) 3/5 ; 2 ; 5

41) The standard electrode potentials $E^o_{I_2/I^-} = 0.54V$, $E^o_{Br_2/Br^-} = +1.09V$, $E^\circ_{Fe^{+2}/Fe} = -0.44V$. On the basis of above data which of the following process is non-spontaneous?

(1) $Br_2 + 2I^- \rightarrow 2Br^- + I_2$

(2) $I_2 + 2Br^- \rightarrow 2I^- + Br_2$

(3) $Fe + Br_2 \rightarrow Fe^{+2} + 2Br^-$

(4) $Fe + I_2 \rightarrow Fe^{+2} + 2I^-$

42) E.M.F. of cell
Ni $|Ni^{2+}$ (1.0M)| $|Au^{3+}$(1.0M)| Au
is .... if E° for Ni^{2+}/Ni is –0.25 V and E° for
Au^{3+}/Au is 1.50 V :-

(1) +1.25 V

(2) –1.75 V

(3) +1.75 V

(4) +4.0 V

43) Oxidation number of Fe in $[Fe(NO)(CN)_5]^{-2}$ is -

(1) +3

(2) +2

(3) –3

(4) –2

44) In which of the following reactions H_2O_2 is acting as a reducing agent ?

(1) $SO_2 + H_2O_2 \rightarrow H_2SO_4$

(2) $2KI + H_2O_2 \rightarrow 2KOH + I_2$

(3) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

(4) $Ag_2O + H_2O_2 \rightarrow 2Ag + H_2O + O_2$

45) In the reaction :
$xBrO_3^- + yCr^{3+} + zH_2O \rightarrow Br_2 + CrO_4^{2-} + H^+$
the coefficients x, y and z are :

(1) 6, 10, 11

(2) 6, 10, 20

(3) 6, 8, 22

(4) 6, 10, 22

BIOLOGY

1) The innermost layer or last layer of cortex is called :-

(1) Pericycle

(2) Conjunctive tissue

(3) Endodermis

(4) Exodermis

2) In a dorsiventral leaf, the location of palisade tissue and phloem, respectively towards :-

(1) adaxial and abaxial

(2) abaxial and adaxial

(3) adaxial and adaxial

(4) abaxial and abaxial

3) The transverse section of a plant shows the vascular bundles are conjoint, scattered and surrounded by a sclerenchymtous bundle sheath. This transvers section shows :–

(1) Monocot root

(2) Dicot root

(3) Monocot stem

(4) Monocot leaf

4) **Statement–1** :- Radial vascular bundles are found in stems and leaves.
Statement–2 :- Exarch primary xylem is found in stems.

(1) Both Statement 1 and Statement 2 are correct.

(2) Both Statement 1 and Statement 2 are incorrect

(3) Statement 1 is correct and Statement 2 is incorrect

(4) Statement 1 is incorrect and Statement 2 is correct

5) **Assertion :** Endodermis is also called as starch sheath in Dicot stem.
Reason : The cells of endodermis are rich in starch grain.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

6) Match List–I and List–II

	List–I		**List–II**
(i)	Dicot Root	(a)	Pith large
(ii)	Monocot Root	(b)	Casparian strips
(iii)	Dicot leaf	(c)	Mesophyll
(iv)	Monocot stem	(d)	Water cavities

(1) (i)–a, (ii)–c, (iii)–b, (iv)–d
(2) (i)–b, (ii)–a, (iii)–c, (iv)–d
(3) (i)–a, (ii)–b, (iii)–c, (iv)–d
(4) (i)–a, (ii)–d, (iii)–c, (iv)–b

7) Vascular bundles in monocotyledonous stem are considered closed because :

(1) There are no vessels with preforation
(2) Cambium is present
(3) Cambium is absent
(4) Xylem surrounded all around by phloem.

8) In leaves, the ground tissue consist of thin walled chloroplast containing cells. such type of cells form ?

(1) Bulliform cells
(2) Mesophyll
(3) Idioblast cells
(4) Pericycle

9) Ground tissue includes :

(1) All tissues external to endodermis
(2) All tissues except epidermis & vascular bundle
(3) Epidermis & cortex
(4) All tissues internal to endodermis

10) In leaves, position of protoxylem :-

(1) Towards the adaxial side
(2) Towards the abaxial side
(3) Surrounded by metaxylem
(4) Scattered in middle

11) Conjoint and open type of vascular bundle are found in :-

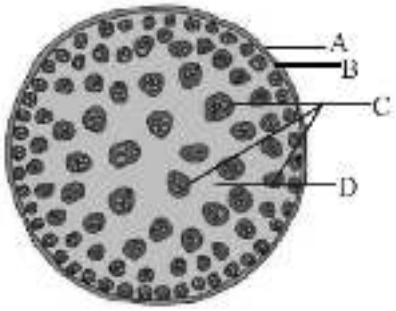
(1) Dicot root
(2) Monocot stem
(3) Dicot stem
(4) Leaves

12)

Which of the following is correct about the monocotyledonous stem?

(1) Sclerenchymatous layers below the epidermis.
(2) Xylem parenchyma is absent
(3) Ground tissue is absent
(4) Parenchymatous bundle sheath

13) Given below is the diagrammatic view of T.S. of stem. Identify the parts labelled A, B, C and D



and select the right option about them.

	A	B	C	D
(1)	Epidermis	Hypodermis	Ground tissue	Vascular bundles
(2)	Epidermis	Hypodermis	Vascular bundles	Ground tissue
(3)	Hypodermis	Epidermis	Ground tissue	Vascular bundles
(4)	Hypodermis	Ground tissue	Epidermis	Vascular bundles

(1) 1
(2) 2
(3) 3
(4) 4

14) Read the following terminology of column I & II and identify their correct match :

	Column-I		Column-II
(A)	Dumb-bell shaped guard cell	(i)	Roots
(B)	Radial vascular bundles	(ii)	barrel-shaped cells
(C)	Phloem parenchyma absent	(iii)	Grasses
(D)	Endodermis	(iv)	Root hairs
		(v)	Monocotyledons stem

(1) A-iii, B-i, C-iv, D-ii
(2) A-iii, B-i, C-ii, D-v
(3) A-iii, B-i, C-v, D-ii
(4) A-iv, B-iii, C-ii, D-i

15) The roots, which originate from the base of the stem in wheat are

(1) Fibrous root
(2) Tap root
(3) Secondary root
(4) Adventitious root

16) Find out the right sequence to various regions of root tip from apex to base

(1) Maturation zone, elongation zone, meristematic zone and root cap
(2) Root cap, meristematic zone, elongation zone and maturation zone
(3) Root cap, elongation zone, meristematic zone and maturation zone
(4) Maturation zone, meristematic zone, elongation zone and root cap

17) Regarding alternate phyllotaxy which of the following is odd

(1) China rose
(2) Mustard
(3) Sunflower
(4) *Alstonia*

18) In some flowers like *Lily*, there is no differentiation in calyx and corolla, which is termed as :-

(1) Florets
(2) Gamopetalous
(3) Perianth
(4) Polypetalous

19) Inferior ovary is found in :-

(1) Ray florets of sunflower

(2) Mustard
(3) China rose
(4) Rose

20) In which type of placentation, the ovules develop on the inner wall of the ovary :-

(1) Marginal
(2) Parietal
(3) Axile
(4) Free central

21) Mango fruit develops from –

(1) Monocarpellary gynoecium only
(2) Apocarpous gynoecium
(3) Superior ovary only
(4) Both (1) & (3)

22) ***Assertion (A) :*** Flowers of solanaceae are actinomorphic.
Reason (R) : Flowers of these families can be cut into two equal halves through any vertical plane passing through the centre.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true and (R) is the not correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

23) Diadelphous stamens are found in :-

(1) Onion
(2) Pea
(3) Wheat
(4) China rose

24)

How many given plants have inferior ovary in flowers :
mustard, china rose, brinjal, plum, rose, guava, cucumber

(1) One
(2) Two
(3) Three
(4) Four

25) The main functions of the root systems are
(a) Absorption of water & minerals
(b) Providing a proper anchorage to the plant

(c) Maintaining the temperature & pressure of the plant
(d) Gaseous exchange

(1) a & b
(2) c & d
(3) a & c
(4) b & d

26) Pinnately compound leaves are found in :-

(1) Silk cotton
(2) Neem
(3) Mango
(4) Peepal

27) Arrangement of flowers in racemose and cymose inflorescence are respectively –

(1) Basipetal, Basipetal
(2) Basipetal, Acropetal
(3) Acropetal, Basipetal
(4) Acropetal, Acropetal

28) Identify the placentation shown below as well as related example of its occurrence, select the option in which both, type of placentation and its related examples are correctly given –



	Type ofPlacentation	**Examples**
(1)	Marginal	Pea and bean
(2)	Parietal	*Argemone* and mustard
(3)	Axile	Onion and garlic
(4)	Basal	Sunflower and marigold

(1) 1
(2) 2
(3) 3
(4) 4

29) Function of petiole is :-

(1) Hold the leaf blade
(2) Allow leaf blades to flutter in wind
(3) Bringing fresh air to leaf surface
(4) All of the above

30) A typical flower has four different kinds of whorls arranged successively on the swollen end of the a stalk or pedicel called :-

(1) Ovary
(2) Carpel
(3) Thalamus
(4) Peduncle

31)

Match the column A and column B :-

Column-A		**Column-B**	
(i)	Lamina incised but incision do not touch the midrib	(a)	Rachis absent
(ii)	Leaflets attached to a common Point	(b)	Pinnately compound
(iii)	Leaflets attached on rachis	(c)	Palmately compound
(iv)	Palmately compound leaf	(d)	Simple leaf

(1) i-a, ii-b, iii-c, iv-d
(2) i-b, ii-c, iii-d, iv-a
(3) i-c, ii-d, iii-b, iv-a
(4) i-d, ii-c, iii-b, iv-a

(1) 1
(2) 2
(3) 3
(4) 4

32) **Assertion (A)** :- Ovary is one-chambered but later on it becomes two-chambered in mustard.
Reason (R) :- False septum is formed in mustard.

(1) Both Assertion and Reason are correct and reason is the correct explanation of Assertion.
(2) Both assertion and reason are true but reason is not the correct explanation of assertion.
(3) A is correct but R is incorrect
(4) A is incorrect but R is correct

33) How many of the given examples have axile type of placentation :-
Pea, Chinarose, tomato, lemon, sunflower, primrose, argemone.

(1) 4

(2) 3

(3) 6

(4) 7

34) Which one of the following options gives the correct categorisation of six plants according to the type of symmetry (A, B, C) they exibit.

	A ACTINOMORPHIC	**B ZYGOMORPHIC**	**C ASYMMETRIC**
(1)	*Datura*, mustard	Pea, bean, gulmohur	Canna
(2)	Pea, bean, gulmohur	*Datura*, mustard	Canna
(3)	Pea, bean, gulmohur	Canna	*Datura*, mustard
(4)	Canna	Pea, bean, gulmohur	*Datura*, mustard

(1) 1

(2) 2

(3) 3

(4) 4

35) Match the Column I with Column II.

	Column-I		**Column-II**
(A)	Edible mesocarp	(i)	Coconut
(B)	Endospermous seed	(ii)	Mango
(C)	Fibrous mesocarp	(iii)	Bean
(D)	Non-endospermous seed	(iv)	Castor
(E)	Ovules	(v)	Future fruit
(F)	Ovary	(vi)	Future seed

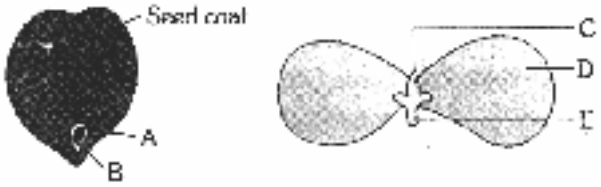
(1) A-(ii), B-(iv), C-(vi), D-(i), E-(iii), F-(v)

(2) A-(i), B-(iii), C-(v), D-(ii), E-(iv), F-(vi)

(3) A-(vi), B-(v), C-(iv), D-(iii), E-(ii), F-(i)

(4) A-(ii), B-(iv), C-(i), D-(iii), E-(vi), F-(v)

36) The following diagram is the typical structure of dicotyledonous seeds. In which one of the



options all the five parts A to E are correctly identified

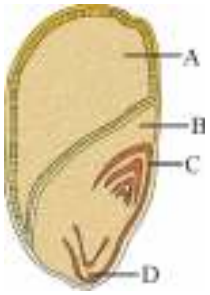
(1) A - Hilum, B - Micropyle, C - Plumule, D - Radicle, E - Cotyledon

(2) A - Micropyle, B - Hilum, C - Plumule, D - Cotyledon, E - Radicle

(3) A - Hilum, B - Micropyle, C - Plumule, D - Cotyledon, E - Radicle

(4) A - Hilum, B - Micropyle, C - Radicle, D - Cotyledon, E - Plumule

37) Given below is the floral diagram of monocotyledons seed identify the parts labelled A, B, C, D.



	Part A	**Part B**	**PartC**	**Part D**
(1)	Endosperm	Scutellum	Coleoptile	coleorhiza
(2)	Scutellum	Endosperm	coleorhiza	Coleoptile
(3)	Scutellum	coleorhiza	Coleoptile	Endosperm
(4)	coleorhiza	Coleoptile	Scutellum	Endosperm

(1) 1

(2) 2

(3) 3

(4) 4

38) In floral formula, (K) denotes :

(1) Polysepalous

(2) Gamosepalous

(3) Polypetalous

(4) Gamopetalous

39) In which of the following character is **not** observed in given floral diagram ?



(1) Tetradynamous androecium

(2) Parietal placentation

(3) Valvate aestivation of petals

(4) Epigynous flower

40) **Assertion :** In *Cassia* and gulmohar the aestivation is called Imbricate.
Reason : One margin of appendage overlaps that of next one but without any particular direction.

(1) Assertion and reason both are true and the reason is correct explanation of assertion

(2) Assertion and reason both are true but reason is not correct explanation of assertion

(3) Assertion is true but reason is wrong

(4) Assertion and reason both are wrong

41) When stamens are attached to the petals they are called ___A___ and stamens are attached to the perianth they are called ___B___ and stamens are free they are called ___C___.
Select correct option for A,B and C and fill in the blanks:-

(1) A → Epiphyllous, B → Epipetalous, C → diadelphous

(2) A → Polyandrous , B → Epiphyllous, C → Epipetalous

(3) A → Monoadelphous, B → Polyandrous, C → Epipetalous

(4) A → Epipetalous, B → Epiphyllous, C → Polyandrous

42) The arrangement of veins & veinlets in leaf lamina is known as-

(1) Phyllotaxy

(2) Venation

(3) Aestivation

(4) Placentation

43) **Statement-I :** In Monocot stem, peripheral vascular bundles are generally smaller than centrally located ones.
Statement-II : In grasses, leaves contain bulliform cells.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II is incorrect.

44) Which of the following is an example of endospermic seed ?

(1) Gram
(2) Pea
(3) Orchid
(4) Castor

45) Casparian strips, that are observed on endodermis are made up of :-

(1) Suberin
(2) Pectin
(3) Hemicellulose
(4) Cellulose

46)

Among which one of the following groups of chemicals, all are neurotransmitters?

(1) glycine, dopamine, melatonin
(2) calcitonin, serotonin, acetylcholine
(3) noradrenaline, calcitonin, thyroxine
(4) acetylcholine, noradrenaline, dopamine

47) Match **List–I** with **List–II**

List–I		**List–II**	
A.	Limbic system	I.	Body temperature
B.	Arachnoid	II.	Tract of nerve fibres
C.	Corpus callosum	III.	Amygdala
D.	Hypothalamus	IV.	Cranial meninges

Choose the **correct** answer from the options given below :
(1) A–I, B–II, C–IV, D–III
(2) A–II, B–I, C–III, D–IV
(3) A–IV, B–II, C–III, D–I
(4) A–III, B–IV, C–II, D–I

48) Which part of the hindbrain is connected to the spinal cord ?

(1) Brain stem
(2) Pons
(3) Cerebellum
(4) Medulla

49) Which of the following contains a number of centres which control body temperature, urge for eating and drinking :

(1) Thalamus

(2) Medulla oblongata

(3) Hypothalamus

(4) Cerebrum

50) Which of the following is a major coordinating centre for sensory and motor signaling :

(1) Cerebrum

(2) Hypothalamus

(3) Corpus callosum

(4) Thalamus

51) Choose the correct option from given below :
(a) Association areas are clearly sensory
(b) Association areas are neither clearly sensory nor motor
(c) Association areas are responsible for complex functions
(d) Association areas are clearly motor

(1) a, c

(2) a, b, c

(3) b, c

(4) a, b, c, d

52) The cerebral cortex is referred to as the __A__ due to its __B__ appearance the neuron __C__ are concentrated here giving the colour :

	A	B	C
(1)	White mater	White	Cell bodies
(2)	Grey mater	Grey	Axon
(3)	White mater	White	Axon
(4)	Grey mater	Grey	Cell bodies

(1) 1

(2) 2

(3) 3

(4) 4

53) Cerebral hemisphares are connected by a tract of nerve fibres called as :

(1) Corpus albicans

(2) Thalamus

(3) Corpus callosum

(4) Cerebellar peduncle

54) A nerve impulse is transmitted from one neuron to another through junction called as ?

(1) Neuro muscular junction

(2) Synapse

(3) Telodendria

(4) Node of Ranvier

55) In resting condition the outer surface of the axonal membrane possesses a __A__ while its inner surface becomes __B__ and therefore is __C__.

	A	B	C
(1)	Positive charge	Negative charge	Depolarised
(2)	Positive charge	Negative charge	Polarised
(3)	Negative charge	Positive charge	Polarised
(4)	Positive charge	Negative charge	Hyper polarised

(1) 1

(2) 2

(3) 3

(4) 4

56) The axoplasm contains high concentration of __A__ and negatively charged proteins and low concentration of __B__ ?

	A	B
(1)	$K^{\oplus}$ ions	PO_4^{-3} ions
(2)	$Na^{\oplus}$ ions	$K^{\oplus}$ ions
(3)	$K^{\oplus}$ ions	$Na^{\oplus}$ ions
(4)	PO_4^{-3} ions	$K^{\oplus}$ ions

(1) 1

(2) 2

(3) 3

(4) 4

57) When a neuron is not conducting any impulse the axonal membrane is comparatively more permeable to __A__ and nearly impermeable to __B__.

	A	B
(1)	$Na^{\oplus}$ ions	$K^{\oplus}$ ions
(2)	$K^{\oplus}$ ions	$Na^{\oplus}$ ions
(3)	PO_4^{-3} ions	$Na^{\oplus}$ ions
(4)	Negative charge proteins	$K^{\oplus}$ ions

(1) 1

(2) 2

(3) 3

(4) 4

58) Ion channels are _____ to different ions.

(1) Semipermeable

(2) Permeable

(3) Selectively permeable

(4) Impermeable

59)

	Column-I		**Column-II**
(A)	Multipolar Neuron	(i)	Retina of eye
(B)	Bipolar Neuron	(ii)	Cerebral cortex
(C)	Unipolar Neuron	(iii)	Cell body
(D)	Nissl's granules	(iv)	Embryonic stage

	A	B	C	D
(1)	ii	iii	i	iv
(2)	ii	i	iii	iv
(3)	ii	iv	i	iii
(4)	ii	i	iv	iii

(1) 1

(2) 2

(3) 3

(4) 4

60) Fibres which transmit impulses towards the cell body called as :

(1) Axon terminal

(2) Axon

(3) Dendrites

(4) Axon hillock

61) The neural system is composed of highly specialised cells which can detect, receive and transmit different kinds of stimuli are called as ?

(1) Nerve

(2) Neuron

(3) Dendritic cells
(4) Telodendria

62) Thyroid gland is :-

(1) Bilobed gland
(2) Endodermal in origin
(3) Largest endocrine gland
(4) All of the above

63) Autonomic neural system transmits impulses from the CNS to -

(1) Involuntry organs and smooth muscles
(2) Voluntry organs and skeletal muscles
(3) Involuntry organs and skeletal muscles
(4) Voluntry organs and smooth muscles

64) **Assertion (A) :-** The autonomic neural system is further classified into sympathetic and parasympathetic neural system.
Reason (R) :- The sensory organs detect all types of changes in the environment and send appropriate signals to the ANS.
Choose the correct choice :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(2) (A) is incorrect but (R) is correct
(3) (A) is correct but (R) is not correct
(4) Both (A) and (R) are correct but (R) is the correct explanation of (A)

65) Which lobular structure is located between lungs, behind sternum on the ventral side of aorta ?

(1) Pineal
(2) Parathyroid
(3) Thyroid
(4) Thymus

66) Peripheral Nervous system does not consist of :

(1) Afferent fibres
(2) Efferent fibres
(3) Somatic Neural system
(4) Brain

67) In central nervous system myelin sheath is produced by __A__ and in Peripheral Nervous system myelin sheath is produced by __B__.
Select the correct option about A and B respectively :-

(1) Oligodendrocytes and Astrocytes

(2) Oligodendrocytes and Schwann cells
(3) Schwann cells and Astrocytes
(4) Schwann cells and Oligodendrocytes.

68) Which of the following statement is not incorrect ?

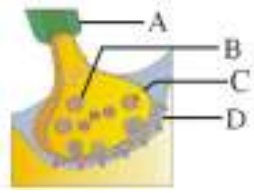
(1) Corpus callosum is a band of fibres, which connect left and fight cerebellar hemisphere
(2) Hypothalamus produces and release hormones and regulate breathing and micturition
(3) Limbic system inter connect different regions of brain and control movement
(4) Medulla oblongata controls respiration and cardiovascular reflexes.

69) When the outside temperature is cold then which part of brain maintain 37ºC temperature of body ?

(1) Cerebrum
(2) Hypothalamus
(3) Corpus callosum
(4) Medulla oblongata

70) The gaps between two adjacent myelin sheaths are called as -

(1) Nissl's granules
(2) Synaptic vesicles
(3) Neurotransmitters
(4) Nodes of ranvier

71) 

In the given diagram, which one is receptor sites for neuro transmitters?

(1) A - Tips of Axons
(2) B - Membrane of Synaptic Vesicles
(3) C - Pre - Synaptic Membrane
(4) D - Post Synaptic Membrane

72) Brain–stem of human brain consist of __A__, pons & medulla oblongata.
Find correct option & fill in the blank space.

(1) Crebrum
(2) Cerebellum
(3) Mid brain
(4) Diencephalon

73) What is the main function of the medulla oblongata?

(1) Coordination of muscle movements

(2) Control of heart beat and respiration

(3) Regulation of body temperature

(4) Memory storage

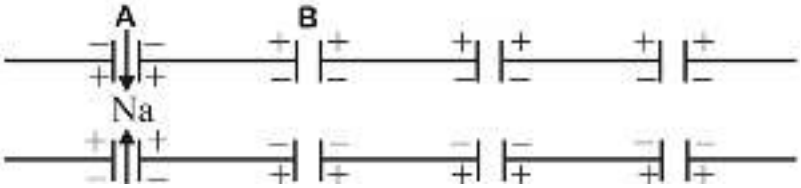
74) The cerebrum wraps around a structure called....., which is a major coordinating centre for sensory and motor signaling

(1) Thalamus

(2) Hypothalamus

(3) Mid brain

(4) Pineal body

75) 

What is the 'A' & 'B' ?

(1) 'A' - depolarisation, 'B' - polarisation

(2) 'A' - polarisation, 'B' - depolarisation

(3) 'A' - depolarisation, 'B' - hyperpolarisation

(4) 'A' - repolarisation, 'B' - polarisation

76) Which of the following disorders occur due to hypersecretion of cortisol ?

(1) Grave's disease

(2) Addison's disease

(3) Cushing's syndrome

(4) Cretinism

77)

Hormones of fight and flight are responsible for :-

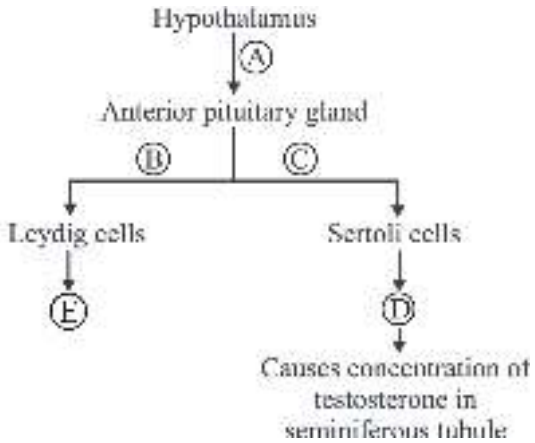
(1) Inhibition of sweating

(2) Pupilary contraction

(3) Piloerection

(4) Increase strength of heart relaxation

78) Observe the following flow chart and choose right statement :-



(1) B cause release of E and E inhibits secretion of D (normally)

(2) C causes release of D and D concentrates E inside the seminiferous tubule

(3) A is GnRH, C is ICSH and B is FSH

(4) A is GnRH, C is LH and B is FSH

79) Find out the correct statement
(A) Gluconeogenesis & lipolysis are stimulated by glucocorticoids of adrenal.
(B) In diabetes insipidus patients are successfully treated with insulin therapy.
(C) When blood pressure decreases, ANF is secreted which causes dialation of blood vessels to increase B.P.
(D) Glucagon reduces cellular glucose uptake and utilisation.

(1) A, C

(2) B, C

(3) A, D

(4) B, D

80) Amino acid derivative hormones are :-

(1) Glucagon, Adrenaline, Nor adrenaline

(2) Insulin, Nor adrenaline, MSH

(3) Adrenaline, Nor-Adrenaline, Thyroxine

(4) Thyroxine, Nor adrenaline, ACTH

81)

Pineal gland regulates .....
Thyroid gland regulates .....
Adrenal cortex is the source of .....
Read above incomplete sentences and find out **correct** option to complete them :–

(1) BMR, Ca^{+2} metabolism, release of MSH

(2) Metabolism, Ca^{+2} metabolism, release of few amount of sex steroids

(3) Pigmentation, growth, release of growth hormone

(4) Electrolytic balance, BMR, Ca^{+2} metabolism

82) Choose the correct option for pair of hormone which doesn't have antagonist function.

(1) Insulin-Glucagon
(2) Parathormone-Calcitonin
(3) Melatonin-MSH
(4) Thyroxine-Thymosin

83)

Glucocorticoids stimulates :-
(A) Gluconeogenesis (B) Lipolysis

(C) Proteolysis
(1) Only A and B
(2) Only A
(3) Only B
(4) All A, B and C

84) The growing ovarian follicles mainly secretes which hormone?

(1) Estrogen
(2) Progesterone
(3) Androgens
(4) Testosterone

85) Which statements is incorrect regarding hormone ?

(1) Hormones are non-nutrient chemical
(2) Act as intracellular secondary messenger
(3) Produce in trace amount
(4) Hormones are species non specific

86) Melatonin is secreted by :-

(1) Pineal body
(2) Skin
(3) Pituitary gland
(4) Thyroid

87) Which one of the following hormone never reaches to cytoplasm ?

(1) Estrogen
(2) FSH
(3) Progestrone

(4) Testosterone

88) Thymosin is responsible for :-

(1) Raising the blood sugar level
(2) Raising the blood calcium level
(3) Differentiation of T.lymphocytes
(4) Decrease in blood RBC

89) Pregnancy maintaining hormone is

(1) Oestrogen
(2) Androgen
(3) Progesterone
(4) Gastrin

90) Which is not synthesised by pituitary ?

(1) FSH
(2) MSH
(3) Oxytocin
(4) Prolactin

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	1	1	3	1	3	3	2	4	3	4	3	3	4	2	2	1	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	2	2	3	4	1	3	2	2	2	2	2	4	4	2	1	2	2	2	1
Q.	41	42	43	44	45															
A.	3	2	2	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	3	2	3	3	1	3	4	3	3	4	2	3	3	4	2	3	4	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	4	2	3	1	3	4	3	2	2	1	1	4	1	1	1	2	3	4	1
Q.	86	87	88	89	90															
A.	2	3	2	4	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	1	3	2	1	2	3	2	2	1	3	1	2	3	1	2	4	3	1	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	1	2	2	1	2	3	3	4	3	4	1	2	1	4	3	1	2	4	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	1	4	1	4	4	4	3	4	3	4	3	2	2	3	2	3	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	4	1	3	4	4	2	4	2	4	4	3	2	1	1	3	3	2	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	4	4	1	2	1	2	3	3	3										

SOLUTIONS

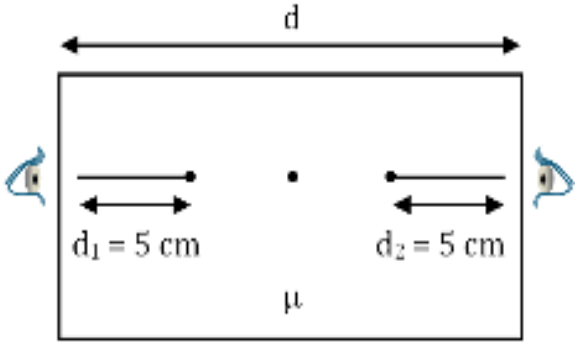
PHYSICS

1) **Explain Question:**
To find the thickness of glass slab (μ = 15) when an air bubble in slab the slab appeared at 5 cm from one face and appeared at 5 cm from opposite side.

Concept:
This question is based on Apparent depth due to refraction

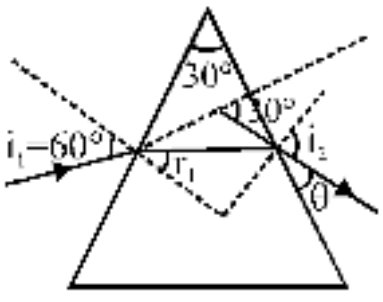
$$d_{app} = \frac{d_{real}}{\mu}$$

Solution:
Given m = 1.5, $(d_1)_{app}$ = 5 cm
$(d_2)_{app}$ = 5 cm
visual aid



$d = \mu(d_1 + d_2)$ given d_1 = 5 cm, d_2 = 5 cm
= 1.5 (5 + 5) = 15 cm

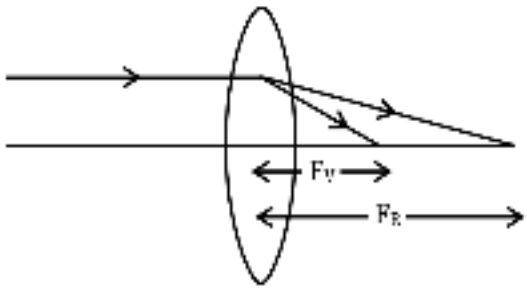
Final Answer:
15 cm
correct option (2)



2)
$\because \quad \delta = i_1 + i_2 - A$
$30° = 60° + i_2 - 30°$
Means, emergent ray goes along the normal & it will make angle with the second face of the prism is $\theta = 90° - 0° = 90°$
$i_2 = 0°$

3) **Explanation:**
To find effect on focal length due to various wavelength for lens.
Concept:
This question is based on lens maker formula and No. of images

Solution:
focal length



$$f = \frac{1}{(\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)}$$

$$\frac{\mu_2}{\mu_1} = \frac{\lambda_1}{\lambda_2}$$

As $\lambda \uparrow \mu \downarrow f \uparrow$

$\lambda_R > \lambda_0 > \lambda_4 > \lambda_G > \lambda_B > \lambda_I > \lambda_V$

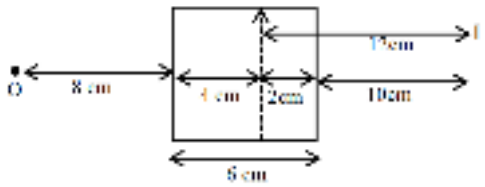
$\mu_R < \mu_0 < \mu_4 < \mu_G < \mu_B < \mu_I < \mu_V$

so $f_R > f_0 > f_4 > f_G > f_B > f_I > f_V$

Final Answer:
Option: (1)

4) Newton's formula,

$f = \sqrt{10 \times 40} = 20$ cm



5)

$$= 6\left(1 - \frac{1}{1.5}\right) = 2\text{cm}$$

6) **Question Explanation:** To find out amph Diameter of Image of moon seen by telescope as per given Quemon.
Concept : Angular magnification by telescope.
Formula : $\frac{\beta}{OC} = \frac{f_o}{f_e}$

angle = $\frac{\text{Are}}{\text{Radicus}}$

Solution :
Angle of moon

$$\alpha = \frac{\text{Diameter of moon}}{\text{Distance from earth}}$$

$$\alpha = \frac{3 \cdot 5 \times 10^3 \times 10^3}{3 \cdot 8 \times 10^5 \times 10^3}$$

$$\Rightarrow \beta = \frac{4}{0 \cdot 1} \times \frac{3 \cdot 5 \times 10^6}{3 \cdot 8 \times 10^8} \times 10$$

= 0.36 Radian

Final Answer : option (1)

7) **1. Explanation of the Question**

A single slit of width 0.20 mm is illuminated with light of wavelength 500 nm. The screen is placed 80 cm away. We need to calculate the width of the central bright fringe.

2. Concept The width of the central bright fringe in single-slit diffraction is given by:

$$W = \frac{2\lambda L}{a}$$

Where:

- λ is the wavelength.
- L is the distance to the screen,
- a is the slit width.

3. Formula

$$W = \frac{2 \times 500 \times 10^{-9} \times 0.8}{0.20 \times 10^{-3}}$$

4. Calculation

$$W = \frac{2 \times 500 \times 10^{-9} \times 0.8}{0.20 \times 10^{-3}} = 4\text{mm}$$

Final Answer: (4 mm).

Hence, the correct answer is option (3).

8) Malus law

$I = I_0\cos^2\theta$

9)

$\mu = \tan i_p$

$= \tan 60$

$= \sqrt{3}$

10) $\lambda_{sound} > \lambda_{light}$

11) Intensity of central maxima is I_0.

% intensity of second maxima

$$\Rightarrow \frac{I_0/61}{I_0} \times 100$$

$$\Rightarrow \frac{100}{61} \simeq 1.64\%$$

12)

$I = I_0\cos^2\theta$ $\quad I_{max.}$ $\theta = 0°, 180°$
$I_{min.}$ $\theta = 90°, 270°$

13) $\lambda = 405 \times 10^{-9}$ m, $a = 4 \times 10^{-6}$ m,
$d = 20 \times 10^{-6}$ m

$$\Rightarrow \beta_d = n\beta i \quad ; \quad \frac{2\lambda D}{a} = n\frac{\lambda D}{d}$$

$$n = \frac{2d}{a} = 10$$

14) After first, $I_1 = \frac{I_0}{2}$
After second, $I_2 = I_1 \cos^2 30°$

$$= \frac{I_0}{2} \times \frac{3}{4} = \frac{3I_0}{8}$$

After third, $I_3 = I_2 \cos^2 30°$

$$= \frac{3I_0}{8} \times \frac{3}{4} = \frac{9I_0}{32}$$

After fourth, $I_3 \cos^2 30°$

$$= \frac{9I_0}{32} \times \frac{3}{4} = \frac{27}{128}I_0 = 0.21\ I_0$$

15)

For the secondary minima $d\sin\theta = n\lambda$
$\text{Sin}\theta = n\lambda/d$
For second minima $n = 2$

$$\sin\theta = \frac{2\lambda}{d} = \tan\theta_1 = \frac{x_1}{D}$$

For fourth minima $n = 4$

$$\sin\theta_2 = \frac{4\lambda}{d} = \frac{x_2}{D}$$

$$x_2 - x_1 = \frac{4\lambda}{d} - \frac{2\lambda}{d} = \frac{2\lambda}{d} = 6 - 3 = 3\text{cm}$$

Width of the central max = 3 cm

16) Motion of planet around sun repeats at regular intervals of time, but it is not a to & fro motion.

17) Due to restoring force, acceleration in SHM is always directed towards the mean position and at extreme position the body momentary comes to rest, but son does not explain assertion

18) $B_0 = \frac{\mu_0 i}{2\pi(2a)} + \frac{2\sqrt{2}\mu_0 i}{\pi(2a)}$

$= \frac{\mu_0 i}{2\pi a}\left[\frac{1}{2} + 2\sqrt{2}\right] \otimes$

19) A is feebly repelled ⇒ A is diamagnetic
B is feebly attracted ⇒ B is paramagnetic
C is strongly attracted ⇒ C is ferromagnetic
D remains unaffected ⇒ D is non-magnetic

20)

U = –MBcosθ
$\theta\uparrow_{es}$ $U\theta\uparrow_{es}$

21) $T_{Sum} = 2\pi\sqrt{\frac{(I_1 + I_2)}{(M_1 + M_2)B_H}}$

$T_{diff} = 2\pi\sqrt{\frac{I_1 + I_2}{(M_1 - M_2)B_H}}$

$\Rightarrow \frac{T_s}{T_d} = \frac{T_1}{T_2} = \sqrt{\frac{M_1 - M_2}{M_1 + M_2}} = \sqrt{\frac{2M - M}{2M + M}} = \frac{1}{\sqrt{3}}$

22) $\tau = MB_H \sin\theta \Rightarrow 0.032 = M \times 0.16 \times \sin 30^\circ$

$\Rightarrow M = 0.4\frac{Nm}{T}$

23) $B = \frac{\mu_0 i}{2R}$

$M = \pi R^2 i$

$\Rightarrow M = \frac{2\pi BR^3}{\mu_0}$

24)

$F_m = Bi\ell \sin\theta$

25)

Magnetic field at centre

$B = \frac{\mu_0 I}{2R}$

$I = \frac{B(2R)}{\mu_0}$

$$I = \frac{(12.56)(2)(5.2 \times 10^{-11})}{4\pi \times 10^{-7}}$$
$I = 1.04 \times 10^{-3}$ A

26) Using eE = evB
$\Rightarrow E = vB = 5 \times 10^6 \times 0.02 = 10^5\ Vm^{-1}$

27) $\oint \vec{B} . \vec{dl} = \mu_o I_{in}$
Here : $I_{in} = I_2 - I_1$



28) $B = \frac{\mu_0 Ni}{2\pi r}$
$\frac{B_1}{B_2} = \frac{N_1}{N_2}\frac{r_2}{r_1} = \left(\frac{200}{100}\right)\left(\frac{80}{40}\right) = 4 : 1$

29) $P_1 + \rho gh + 0 = P_2 + \frac{1}{2}\rho v^2$
$v^2 = \frac{2\Delta P}{\rho} + 2gh$
$= \frac{2(4.3-1) \times 10^5}{660} + 2 \times 10 \times 75$
$v^2 = 2500$
v = 50 m/s

30) $Q_1 = Q_2$
$\alpha = A\sqrt{2gh}$
$h = \frac{\alpha^2}{2gA^2}$

31)

$R = 2\sqrt{h(H-h)} = 2\sqrt{\frac{H}{4}\left(H - \frac{H}{4}\right)} = \frac{\sqrt{3}H}{2}$

32) $P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$
$\Rightarrow P + \frac{1}{2}\rho v^2 = \frac{3P}{4} + \frac{1}{2}\rho v_2^2$
$\Rightarrow \frac{1}{2}\rho v_2^2 = \frac{P}{4} + \frac{1}{2}\rho v^2$

$$\Rightarrow v_2^2 = \frac{P}{2\rho} + v^2$$

$$\Rightarrow v_2 = \sqrt{v^2 + \frac{P}{2\rho}}$$

33)

$V = \frac{2}{9\eta} r^2 (\rho - \sigma)g$

Here $v_1 = v_2$

$$r_1^2(\rho_1 - \sigma) = r_2^2(\rho_2 - \sigma)$$

$$\frac{r_1^2}{r_2} = \frac{11-2}{8-2} = \frac{3}{2} \Rightarrow \frac{r_1}{r_2} = \sqrt{\frac{3}{2}}$$

34) $V_T \propto r^2$

$$\frac{r_1}{r_2} = \left(\frac{V_{T_1}}{V_{T_2}}\right)^{\frac{1}{2}} = \left(\frac{9}{4}\right)^{\frac{1}{2}} = \frac{3}{2}$$

Vol. $\propto r^3 \Rightarrow \frac{(Vol.)_1}{(Vol.)_2} = \left(\frac{3}{2}\right)^3 = \frac{27}{8}$

35)

$$F = \eta A \frac{dv}{dx} = 0.07 \times 0.1 \times \frac{1}{10^{-3}}$$
$$= 7 \text{ N}$$

36)

Generated by Allie

Problem Statement: The question presents an assertion and a reason about why a thin stainless steel needle can float on still water. It asks us to evaluate if the assertion and reason are true, and whether the reason correctly explains the assertion.

Underlying Concept: The core concept involves forces acting on objects placed on a water surface. Typically, objects float if the buoyant force balances their weight (Archimedes' principle). However, a thin needle floats because surface tension-the cohesive force at the water surface-supports its weight without it sinking, rather than buoyancy.

Tips and Tricks: Remember that buoyancy applies when an object displaces fluid, but surface tension can support small objects if they don't break the water surface. This distinction helps in cases involving floating needles or insects like water striders.

Common Mistakes: Assuming all floating objects depend solely on buoyancy forces. Another mistake is thinking surface tension is a form of buoyancy when it is a separate force related to intermolecular attraction at the liquid surface.

Why Other Options Are Incorrect?: Answers stating that the reason (buoyancy) explains why the needle floats are incorrect because the needle floats due to surface tension, not buoyancy. Answers denying the assertion are wrong since a needle can float if placed carefully without breaking the water surface.

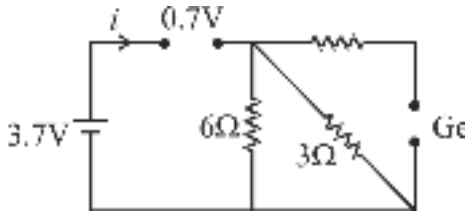
37)

$F = 4\pi RT$
$= 4 \times 3.14 \times 2 \times 10^{-2} \times 3 \times 10^{-2}$
$= 7.5 \times 10^{-3}$ N
$= 7.5$ m N

38) In case of soap bubble,
$W = T \times 2 \times \Delta A$
$= 0.03 \times 2 \times 40 \times 10^{-4} = 2.4 \times 10^{-4}$ J.

39) $P_{excess} = \dfrac{4T}{r}$ $\qquad d = 1cm,\ r = \dfrac{d}{2}$
$= \dfrac{4 \times 25 \times 10^{-3}}{\frac{1}{2} \times 10^{-2}}$
$= 20\ N/m^2$ or 20 Pascal

40) Photo diode detect light when
$\Delta Eg \leqslant \dfrac{hc}{\lambda}$
$\Delta Eg \leqslant \dfrac{12400}{6000}$
$\Delta Eg \leqslant 2.06\,eV$

41)


$R_{eq} = 2\ \Omega$
$i = \dfrac{3.7 - 0.7}{2} = \dfrac{3}{2} = 1.5$ Amp

42) $I_{max} = \dfrac{P_{max}}{V_D} = \dfrac{100 \times 10^{-3}}{0.5} = 200 \times 10^{-3} A$
$R = \dfrac{V - V_D}{I_{max}} = \dfrac{1.5 - 0.5}{200 \times 10^{-3}} = 5\Omega$

43)

For intrinsic semiconducutor,

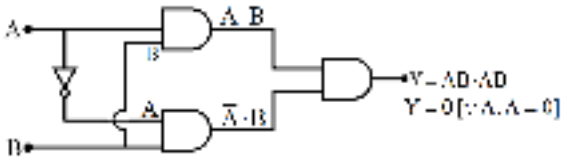
$n_e = n_h = n_i = 7.07 \times 10^{15}\ m^{-3}$

In extrinsic (doped) semiconductor,

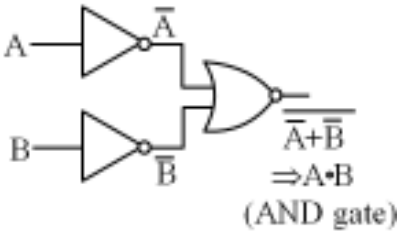
$n_e\, n_h = n_i^2$

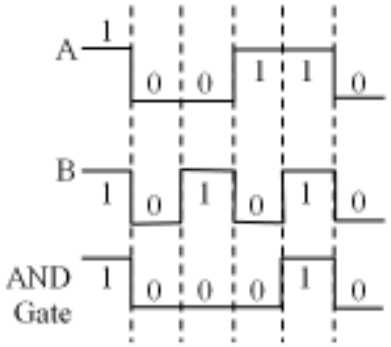
$\Rightarrow \quad n_e = \frac{n_i^2}{n_h} = \frac{(7.07 \times 10^{15})^2}{5 \times 10^{22}} = 10^9\ m^{-3}$

44)



45)





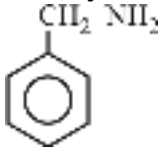
CHEMISTRY

46)

Question Explanation: Identifying the strongest base.

Concept: Basicity

Solution: The basicity of an amine is determined by the availability of the lone pair on the nitrogen atom. The lone pair is more available in aliphatic amines than in aromatic amines. The compound in option A, benzylamine, is the strongest base.

Duo to localized lone pair 

is the strongest base

47)

Due to –M effect of NO_2.

48)

Fact based

49)

'N' never be pentavalent.

50)

Follow Huckel's Rule.

51)

Due to Resonance & Aromaticity stability increases.

52)

No resonance of lone pair (localised □.p.)

53)

Fact based

54)

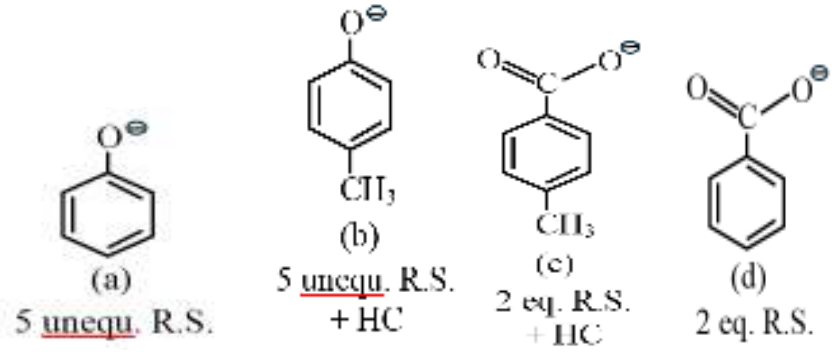
Fact based

55) **Question Explanation :**
This question is asking about acidic strength compassion.

Concept :

This question is based on $\text{Acidic strangth} \propto \frac{-I;\ -M}{+I;\ +M;\ +HC} \propto \text{Stability of conjugated base}$

Solution :



K_a = d > c > a > b

Final Answer :
Option (3)

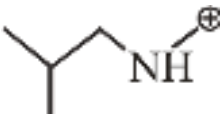
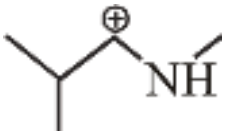
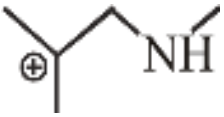
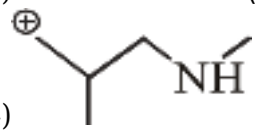
56)

Symmetric molecule has zero dipole moment

57) **Question Explanation :**
Question is asking about most stable carbocation.

Concept :
This question is based on Stability of carbocation α Back bonding (complete octet)

$$\propto \frac{+M; +I; +HC}{-M; -I}$$

Solution :
(1)  (Complete octet pπ-pπ back bonding)
(2)  (Complete octet pπ-pπ back bonding) (α-C-H = 1)
(3)  (α-C-H = 7)
(4)  (α-C-H = 1)
Overall stability of carbocation = (2) > (1) > (3) > (4)

Final Answer :
Option (2)

58)

Fact based

59)

Complete octet species more stable.

60)

Fact based

61)

Due to solvation of □.p.

62)

Fact

63)

α depend on volume but K_P does not depend on volume

64) $O_2(g) \rightleftharpoons CO_2(g)$,
n_o : 4 3
n_e : 4+x 3–x (□ $Q_p > K_p$)
□ = Kp = $\frac{3-x}{4+x}$ Kp = $\frac{1}{5}$

65)

(i) on P↑ Δng = – ve
(ii) T↑ enthermic

66)

$\Delta ng = (n_P - n_R)$
$K_p = K_c(RT)^{\Delta ng}$

67) $K_{eq} = \frac{(P)^{S.C.}}{(R)^{S.C}}$

68)

(i) rex^n is reversed
(ii) rex × 2

69) $K_p = \frac{4\alpha^2}{1-\alpha^2}P$
$K_P - K_P\,\alpha^2 = 4\alpha^2 P$
$K_P = K_P\,\alpha^2 + 4\alpha^2 P$
$K_P = \alpha^2(K_P + 4P)$
$\alpha = \frac{K_P}{K_P + 4P}$
$\alpha = \frac{K_P/P}{\frac{K_P}{P} + 4}$

70)

Explain Question : Compare given equilibrium constants for oxide → elements reaction to determine most stable oxide.

Concept : This question is based on Stability of reactant $\propto 1/K_{(c)}$

Solution :
Given K values:
Option 1: $K = 6.7 \times 10^{16}$
Option 2: $K = 2.2 \times 10^{30}$
Option 3: $K = 1.2 \times 10^{34}$
Option 4: $K = 3.5 \times 10^{33}$
Smallest $K = 6.7 \times 10^{16}$ → highest stability.

Final Answer : option (1)

71) NCERT XI, Part-I, Pg # 195

72)

$A + 2B \rightleftharpoons C \qquad K_1 = \dfrac{[C]}{[B]^2[A]}$

$C + D \rightleftharpoons 3A \qquad K_2 = \dfrac{[A]^3}{[C][D]}$

$6B + D \rightleftharpoons 2C \qquad K_3 = \dfrac{[C]^2}{[D][B]^6}$

$K_1^3 \times K_2 = \dfrac{[C]^3}{[B]^6[A]^3} \cdot \dfrac{[A]^3}{[C][D]} = \dfrac{[C]^2}{[B]^6[D]}$

$\boxed{K_1^3 \cdot K_2 = K_3}$

73)

$K_P = 4P^3$
$P_T = 3P$

74) ΔH = +Ve, favour high temperature .
$\Delta n_g = 0$ No effect of pressure
High concentration of N_2 and O_2

75)

$XY_2(g) \rightleftharpoons XY(g) + Y(g)$
600 – X X X
600 + X = 800
X = 200

$$K_P = \frac{200 \times 200}{400} = 100mm$$

76)

For weak electrolytes

Debye huckle equation = $\Lambda_m^c = A - B\sqrt{C}$

$y = c - mx$

For strong electrolytes

order of condectance HCl > KCl> LiCl

77)

Which have –ve SRP can replace H_2 from acid solution

78)

Ni is attackable electrode so no change in conc^n.

79) $n_{O_2} \times 4 = \frac{965 \times 3}{96500} = 0.03$

$n_{O_2} = \frac{0.03}{4}$

$v_{O_2} = \frac{0.03}{4} \times 22.4$

$= 0.168$ L

80) diameter = 2r

$r = \frac{1}{2} \Rightarrow 0.5$ cm

Area (A) = πr^2

$\Rightarrow 3.14 \times (0.5)^2$

$\Rightarrow 0.785\ cm^2$

Molar conductivity (Λ_m) = $\frac{\kappa \times 1000}{M}$

$(\Lambda_m) = \frac{\left(\frac{\ell}{A} \times \frac{1}{R}\right) \times 1000}{M}$

$= \frac{\frac{50}{0.785} \times \frac{1}{5.55 \times 10^3} \times 1000}{0.05}$

$\Rightarrow 229.65\ cm^2\ mol\ eq^{-1}$

81)

Amount ion discharge depend on charge passed from solution

82) $K_a = \frac{C\alpha^2}{1-\alpha}$

$\alpha = \frac{\lambda_m}{\lambda_m^o}$

83) $\lambda_m^o = (\lambda_m^o)_{2H^+} + (\lambda_m^o)_{CO_3^{2-}}$

84)

R.A. power $\propto \frac{1}{SRP}$

RA power.

X > W > Y > Z

85) $\overset{+2}{FeSO_4} + K\overset{+7}{M}nO_4 \rightarrow Mn^{+2} + Fe^{+3}$

$n_f = 1$ $n_f = 5$

x mol 1 mol

$x \times 1 = 1 \times 5 \Rightarrow x = 5$ mol

$H_2\overset{+3}{C_2}O_4 + K\overset{+7}{M}nO_4 \rightarrow \overset{+2}{Mn}^{+2} + \overset{+4}{C}O_2$

$n_f = 2$ $n_f = 5$

x mol 1 mol

$x \times 2 = 1 \times 5 \Rightarrow x = 5/2$ mol

$\overset{+2}{Fe}\overset{+3}{C_2}O_4 + K\overset{+7}{M}nO_4 \rightarrow \overset{+2}{Mn}^{+2} + \overset{+3}{Fe}^{+3} + \overset{+4}{C}O_2$

$n_f = 2$ $n_f = 5$

x mol 1 mol

$x \times 3 = 1 \times 5$

$x = \frac{5}{3}$ mol

$5 : \frac{5}{2} : \frac{5}{3}$

86)

Since $E^o_{Br_2/Br^-} > E^o_{I_2/I^-}$

 Br_2 gets reduced and I^- gets oxidised.

87)

Cell reaction is -

$3Ni + 2Au^{+3} \rightarrow 3Ni^{+2} + 2Au$

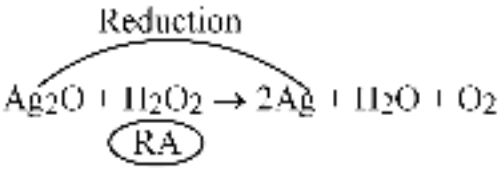
$E^o_{cell} = E^o_{Au^{+3}/Au} - E^o_{Ni^{+2}/Ni}$

$= 1.5 - (-0.25) = 1.75$ V

and $E_{cell} = 1.75 - \frac{0.06}{6}\log_{10}\frac{(1.0)^3}{(1.0)^2} = 1.75V$

88)

O.N. of NO in Fe complex is + 1

89) 

90) $6BrO_3^{\ominus} + 10Cr^{3+} + 22H_2O \rightarrow 3Br_2 + 10CrO_4^{2-} + 44H^{\oplus}$

BIOLOGY

91)

NCERT-XI, Pg. No. # 74

92)

NCERT-XI, Pg. No. # 74

93) NCERT-XI, Pg. No. # 76

94)

NCERT-XI, Pg. No. # 73

95) NCERT-XI, Pg. No. # 75

96) NCERT-XI, Pg. No. # 73-76

97) NCERT-XI, Pg. No. # 76

98) NCERT-XI, Pg. No. # 72

99) NCERT-XI, Pg. No. # 72

100)

NCERT-XI, Pg. No. # 77

101)

NCERT-XI, Pg. No. # 75

102)

NCERT-XI, Pg. No. # 76

103)

NCERT-XI, Pg. No. # 76

104) NCERT-XI, Pg. No. # 72-76

105)

NCERT-XI, Pg. No. # 58

106)

NCERT-XI, Pg. No. # 59

107)

NCERT-XI, Pg. No. # 61

108)

NCERT-XI, Pg. No. # 62

109)

NCERT-XI, Pg. No. # 63

110)

NCERT-XI, Pg. No. # 65

111)

NCERT-XI, Pg. No. # 65

112)

NCERT-XI, Pg. No. # 68

113)

NCERT-XI, Pg. No. # 64

114)

NCERT-XI, Pg. No. # 63

115)

NCERT-XI, Pg. No. # 58

116)

NCERT-XI, Pg. No. # 61

117)

NCERT-XI, Pg. No. # 61

118)

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119) NCERT-XI, Pg. No. # 60

120) NCERT-XI, Pg. No. # 62

121)

NCERT-XI, Pg. No. # 60, 61

122)

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123) NCERT-XI, Pg. No. # 65

124) NCERT-XI, Pg. No. # 62

125)

NCERT-XI, Pg. No. # 66, 67

126)

NCERT-XI, Pg. No. # 66

127)

NCERT-XI, Pg. No. # 67

128)

NCERT-XI, Pg. No. # 67

129)

NCERT-XI, Pg. No. # 67

130)

NCERT-XI, Pg. No. # 64

131) NCERT-XI, Pg. No. # 64

132)

NCERT-XI, Pg. No. # 60

133) NCERT-XI, Pg. No. # 76, 77

134)

NCERT-XI, Pg. No. # 66, 67

135)

NCERT-XI, Pg. No. # 74

136)

Question Explanation:

Identifying a group of neurotransmitters.

Concept:

Neurotransmitters.

Solution:

Neurotransmitters are chemicals that transmit signals across a synapse from one neuron to another. Acetylcholine, noradrenaline, and dopamine are all well-known neurotransmitters. Glycine is also a neurotransmitter, but melatonin is a hormone. Threonine is an amino acid, and somatostatin is a hormone.

Final Answer:

acetylcholine, noradrenaline, dopamine.

137) NCERT-XI, Pg. # 235, 236

138)

Question Explanation:

Identifying the connection between brain and spinal cord.

Concept:

Hindbrain Anatomy.

Solution:

The hindbrain consists of the pons, cerebellum, and medulla oblongata. The medulla oblongata is the lower-most part of the brainstem and is directly connected to the spinal cord. It controls vital functions like breathing and heart rate.

Final Answer:

Medulla.

139) NCERT, Pg # 321(E), 321(H)

140) NCERT, Pg # 321(E), 321(H)

141) NCERT XI Pg.# 236

142) NCERT, Pg # 321(E), 321(H)

143) NCERT, Pg # 321(E), 320(H)

144) NCERT, Pg # 319(E), 318(H)

145) NCERT, Pg # 318(E), 318(H)

146) NCERT, Pg # 317(E), 317(H)

147) NCERT, Pg # 317(E), 317(H)

148) NCERT, Pg # 317(E), 317(H)

149) NCERT, Pg # 317(E), 316(H)

150) NCERT, Pg # 316(E), 316(H)

151) NCERT, Pg # 316(E), 315(H)

152) NCERT-XI, Pg. # 242

153)

Question Explanation: Function of the autonomic nervous system.

Concept: Autonomic Nervous System

Solution:The **autonomic nervous system (ANS)** is part of the peripheral nervous system and is responsible for controlling involuntary actions. It transmits signals from the **central nervous system (CNS)** to internal **involuntary organs** (like the heart, stomach, and liver) and **smooth muscles** to regulate functions such as digestion, heart rate, and blood pressure, without conscious control.

Final Answer: Involuntary organs and smooth muscles

154) NCERT, Pg # 231

155) NCERT Pg.# 243

156)

Question Explanation:

Identifying components not in PNS.

Concept:

Divisions of nervous system.

Solution:

The peripheral nervous system (PNS) consists of all the nerves and ganglia outside the central nervous system (CNS). The brain, being the main part of the CNS, is not a component of the PNS.

Final Answer:

Brain

157) NCERT XII Pg # 232 (E)

158)

Question Explanation:

Identifying the correct statement about brain function.

Concept:

Brain stem and its functions.

Solution:

The correct statement is that the medulla oblongata controls respiration and cardiovascular reflexes. It is a vital center for these involuntary functions. Option 1 is incorrect because the corpus callosum connects the cerebral hemispheres, not the cerebellar hemispheres. Option 2 is wrong because the hypothalamus regulates thirst, hunger, and body temperature but not breathing, and while it produces hormones, it releases some from the pituitary. Option 3 is incorrect because the medulla controls involuntary, not voluntary, movements.

Final Answer:

Medulla oblongata controls respiration and cardiovascular reflexes.

159) NCERT Pg. # 236

160) NCERT Pg. No. # 317

161) NCERT Pg. No. # 319

162) The **brainstem** is a key part of the brain that connects the cerebrum to the spinal cord. It is made up of three main structures: the **midbrain**, the **pons**, and the **medulla oblongata**. These parts are crucial for regulating vital involuntary functions.

163)

The **medulla oblongata** is a vital part of the brainstem. Its primary function is to control

essential involuntary functions necessary for life, such as **regulating heart rate** and **breathing (respiration)**. It also controls other reflexes like sneezing, coughing, and vomiting.

164)

NCERT XI Pg. # 236

165) NCERT Pg. # 318

166)

Solution/Explanation/Calculation:
The disorder that occurs due to hypersecretion of cortisol is **Cushing's syndrome**.

The correct answer is **3. Cushing's syndrome**.

167) **Explain Question:**
Hormones of fight and flight are responsible for :-

Concept :
This question is based on Adrenal medulla hormones.

Explanation : Fight and flight response involves adrenaline/noradrenaline, which cause increased heart rate, blood pressure, sweating, pupillary dilation, and piloerection.
Final answer : 3

168) Module–1

169)

NCERT-XI, Pg. # 242, 244

170) NCERT 337, 338

171) NCERT 333, 335

172)

Generated by Allie

Problem Statement: The question asks to identify which pair of hormones listed does not have opposite (antagonistic) functions. In other words, the pair chosen should not work against each other to regulate a physiological process.

Underlying Concept: Hormones often work in pairs that have opposite effects to maintain

balance in the body, known as antagonistic hormones. For example, insulin lowers blood glucose levels, while glucagon raises them. Understanding which hormones have antagonistic roles helps in identifying how body processes are regulated.

Tips and Tricks: Remember classic hormone pairs like insulin and glucagon or parathormone and calcitonin, which control blood calcium and glucose levels oppositely. If the pair does not involve opposing physiological actions, they are not antagonists.

Common Mistakes: Assuming all hormone pairs listed are antagonistic without reviewing their actual roles. Some hormones do not oppose each other but have independent functions.

Why Other Options Are Incorrect?: - Insulin-Glucagon: They have antagonistic roles in blood glucose regulation. - Parathormone-Calcitonin: They have opposite effects on calcium levels in blood. - Melatonin-MSH: Melatonin regulates sleep-wake cycles, MSH influences skin pigmentation, so they do not oppose each other directly, but they are unrelated rather than antagonistic. - Thyroxine-Thymosin: Thyroxine regulates metabolism, while thymosin is involved in immune system development. They do not act antagonistically.

173) NCERT Pg. # 335, 22.2.7

174) NCERT Pg. # 337, 22.2.10

175)

NCERT Pg. # 330, 22.1

176) NCERT (XI) Pg. # 333 para(22.2.3)

177) NCERT (XI) Pg. # 338 para(22.4)

178) NCERT (XI) Pg. # 334 para(22.2.6)

179)

NCERT XI Pg. # 339

180)

NCERT (XIth) Pg. # 333